

EEE4114F: Problem Set 2 (DTFT)

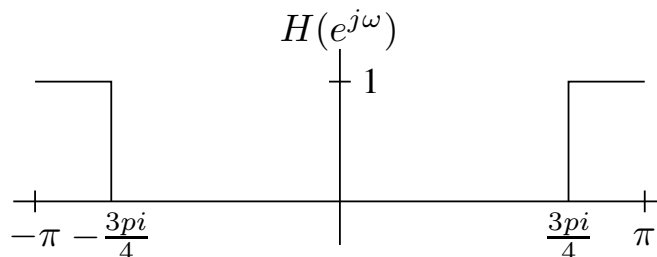
- Find the DTFT of the two-sided sequence $x[n] = (\frac{1}{4})^{|n|}$.
- Let $h[n]$ be the unit sample response of a LTI system. Find the frequency response when
 - $h[n] = \delta[n] + 6\delta[n-1] + 3\delta[n-2]$.
 - $h[n] = (\frac{1}{3})^{n+2} u[n-2]$.
- The input to a LTI system is

$$x[n] = 2 \cos(n\pi/4) + 3 \sin(3n\pi/4 + \pi/8).$$

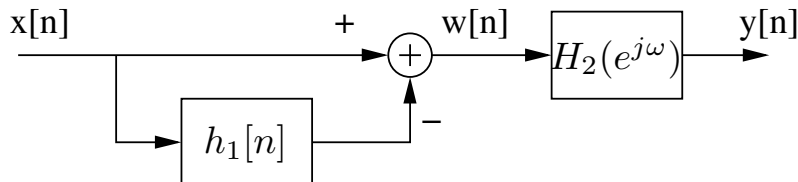
Find the output if the unit sample response of the system is

$$h[n] = 2 \frac{\sin[(n-1)\pi/2]}{(n-1)\pi}.$$

- Consider the highpass filter with cutoff frequency $\omega_c = 3\pi/4$:



- Find the impulse response $h[n]$.
 - A new system is defined so that its unit impulse response is $h_1[n] = h[2n]$. Sketch the frequency response $H_1(e^{j\omega})$ of this system.
- Consider the following interconnection of LTI systems:



where $h_1[n] = \delta[n - 1]$ and

$$H_2(e^{j\omega}) = \begin{cases} 1 & |\omega| \leq \pi/2 \\ 0 & \pi/2 < |\omega| \leq \pi. \end{cases}$$

Find the frequency response and the unit impulse response of the system.

6. Find the DTFT of the following sequences:

(a) $x_1[n] = \left(\frac{1}{2}\right)^n u[n + 3]$.

(b) $x_2[n] = \alpha^n \sin(n\omega_0)u[n]$.

(c)

$$x_3[n] = \begin{cases} (1/2)^n & n = 0, 2, 4, \dots \\ 0 & \text{otherwise.} \end{cases}$$

7. Let $x[n]$ be a sequence with DTFT $X(e^{j\omega})$. For each of the following sequences that are formed from $x[n]$, express the DTFT in terms of $X(e^{j\omega})$:

(a) $x^*[-n]$

(b) $x[n] * x^*[-n]$.

(c) $x[2n + 1]$.

8. Find the frequency response of the LTI system whose input and output satisfies the difference equation

$$y[n] - 0.5y[n - 1] = x[n] + 2x[n - 1] + x[n - 2].$$